

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD organic/materials chemist with 6+ years in synthesis-informed materials characterization, DOE-guided process development, scale-up documentation, and analytical troubleshooting. Connects organic chemistry, LC-MS/NMR/GPC, spectroscopy, thermal analysis, and process controls to define reproducible chemistry and material-quality outcomes.

CORE COMPETENCIES

Organic Process Development

Route / Step Optimization

Scale-Up Readiness

HPLC / LC-MS / NMR

DOE / SPC

Technical Reports

SOPs & Safety

Cross-Functional CMC

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, identifying critical process parameters and translating material-quality data into reproducible process windows and scale-up specifications.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed, synthesized, and characterized organic molecular materials across 15+ variants, connecting structure, purity, stability, and performance through NMR/GPC, LC-MS, spectroscopy, thermal analysis, and electrochemistry.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Drove process development for AFOSR-funded molecular thin-film coatings, optimizing surface preparation and CVD conditions to control morphology, interfaces, and optical performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition with DOE and in-situ QCM-D monitoring, defining growth kinetics and process-control parameters for reproducible coatings.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed process-development and characterization lab operations for a 15-member group; implemented calibration, SOP, safety, and resource-planning systems that improved instrument uptime by 30%.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

DOE-Guided Process Development Current

Built process windows and material-quality specifications for reproducible bio-based polymer scale-up.

DOE, SPC, process parameters, technical reports

Organic Materials Structure-Property Mapping Chemistry of Materials 2025

Connected synthetic design, purity, stability, and performance across xanthene-based organic materials.

Organic synthesis, NMR, LC-MS, GPC, FT-IR, UV-Vis-NIR

AFOSR Thin-Film Scale-Up Controls DoD Funded

Established process-property relationships for CVD/surface-preparation workflows and reproducible thin-film quality.

CVD, QCM-D, surface prep, thin films

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Organic & Process Chemistry: Organic synthesis, route/step optimization, purification, process windows, scale-up documentation, SOPs

Analytical Characterization: LC-MS, GC-MS, NMR, GPC/SEC, FT-IR, UV-Vis-NIR, TGA, DSC, QCM-D, cyclic voltammetry

Process & Documentation: DOE, SPC, JMP (basic), Python (basic), Excel, protocols, technical reports, lab safety and quality systems

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.

3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)